Interacting Branching Gaussian Processes to Quantify Spatio-temporal Cellular Interactions

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Introduction

In cellular biology, Gaussian Processes (GPs) are used to estimate the underlying gene expression levels of cells and model their evolution through time [2]. Estimating gene expression levels of a single cell with a GP is straightforward; however, in real environments, cells divide, interact with each other, and interact with their environment, leading to significant spatial and temporal correlations between the gene expression levels of cells [4]. For gene level modeling, quantifying these interaction terms is often critical for determining the efficacy of treatments targeting these genes. In this project, I propose to model gene expression levels via a branching Gaussian Process model, in which a Gaussian Process evolves and, at a certain time, divides into two GPs. To do so, I treat the collection of the cells' gene expression levels as a multivariate, multi-index Gaussian Process over space and time. Afterwards, I demonstrate with a simulation how spatial correlations can persist after cell division with this model.

Literature Review

There has been extensive research into Gaussian Processes, particularly in cellular biology. Boukouvalas et al. developed the Branching Gaussian Process model (BGP) to determine a posterior distribution of branching times of a population's gene expression levels, which did not address the effect of cell reproduction [1]. Further, the kernel for the BGP assumes that the now distinct cell population regions do not interact [1]. Explicitly, Boukouvalas et al. define three GPs, where f and g are the branches, and h is the trunk:

$$f \sim \mathcal{GP}(0, K) \qquad g \sim \mathcal{GP}(0, K) \quad h \sim \mathcal{GP}(0, K)$$

with the boundary condition at time t_p

$$f(t_p) = g(t_p) = h(t_p)$$

to produce the kernel for f, g, assuming that $\mathbb{E}[\mathrm{Cov}(f(t_1), g(t_2)|f(t_p) = g(t_p))] = 0$

$$\Sigma = \begin{pmatrix} K_{ff} & K_{fg} \\ K_{gf} & K_{gg} \end{pmatrix} = \begin{pmatrix} K(\mathbf{T}, \mathbf{T}) & \frac{K(\mathbf{T}, t_p)K(t_p, \mathbf{T})}{K(t_p, t_p)} \\ \frac{K(\mathbf{T}, t_p)K(t_p, \mathbf{T})}{K(t_p, t_p)} & K(\mathbf{T}, \mathbf{T}) \end{pmatrix}$$

Sarkans et al. then expand on the BGP with their Multivariate Branching Gaussian Process (MBGP) [3]. Crucially, the BGP treats each gene's branches as unique to that gene and independent of other genes. For example, the BGP would consider the PRTN3 and CTSG gene expression levels to be independent of each other. Meanwhile, the MBGP treats the branches as gene expression vectors, capable of expressing the branching of the overall gene characteristics, which is far more realistic for stem cell behaviors. This, however, still does not include spatial interaction effects, only temporal ones.

There is existing literature on spatially dependent GPs. Weber et al. developed a nearest-neighbor Gaussian Process [5]

$$\mathbf{y} \sim \mathcal{N}(\mathbf{X}\beta, \tilde{\Sigma}(\theta, \tau^{\mathbf{2}}))$$

where $\mathbf{y} = (y_1, \ldots, y_N)$ represents the values of N sets of a gene at N locations $\mathbf{s} = (\mathbf{s_1}, \ldots, \mathbf{s_N})$; β is some operator on the spatial positions; and $\tilde{\Sigma}(\theta, \tau^{\mathbf{2}})$ is an approximation—for computational purposes—of the true covariance matrix

$$\Sigma(\theta, \tau^2) = C(\theta) + \tau^2 I \qquad C_{ij}(\theta) = \sigma^2 \exp(\frac{-||s_i - s_j||}{l}).$$

This increases the covariance of local points while incorporating some global value of gene expression levels through the operator on $\mathbf{X}$ in the mean.

Clearly, the gap in the literature lies in the combination of these processes. For this project, to model the gene expression levels of interacting, dividing cells with some global influence requires both the multi-indexed, spatially dependent GPs—to model spatial cellular interactions—and multivariate GPs for the gene expression levels of many cells—to model temporal cellular interactions.

Proposed Research Direction

I propose utilizing a multivariate, multi-index GP to model time dependent and spatial dependent interactions. Let $\{\mathbf{G}\}_{\mathbf{s}}$ be a multivariate GP such that

$$\mathbf{G} \in \mathbb{R}^d \qquad s \in \mathbb{R}^{3d}$$

where d is the number of cells with $3d$ indices for the spacetime position of each cell. For this project, I exclusively model the evolution of one gene expression level. Let us assume there is some global expected value of the gene expression level for all of space. Consider the case for a single cell. Then we can either use additive or multiplicative interactions

$$\mathbf{s} = \begin{bmatrix} t \\ x \\ y \end{bmatrix}$$

$$G_{\mathbf{s}} \sim \mathcal{GP}(f(s), K) \qquad K = K_{tt}K_{xx}K_{yy} \quad \text{or} \quad K = K_{tt} + K_{xx} + K_{yy}, \tag{1}$$

for which I will assume

$$K_{tt} = K(t, t') = \sigma_t^2 \exp(-\gamma_t |t - t'|)$$
$$K_{xx} = K(x, x') = \sigma_x^2 \exp(-\gamma_x |x - x'|)$$
$$K_{yy} = K(y, y') = \sigma_y^2 \exp(-\gamma_y |y - y'|).$$

I will abbreviate the single cell case as

$$G_{\mathbf{s}} \sim \mathcal{GP}(f(\mathbf{s}), K) \qquad K = K(\mathbf{s}, \mathbf{s}').$$

Now I will incorporate the findings from [1] for two cells, assuming a branching time and position $\mathbf{s_b}$:

$$\begin{bmatrix} G_1 \\ G_2 \end{bmatrix}_{s_1, s_2} \sim \mathcal{GP}(f(\mathbf{s_1}, \mathbf{s_2}), \mathbf{K})$$

where

$$\mathbf{K} = \begin{bmatrix} K_{G_1G_1} & K_{G_1G_2} \\ K_{G_2G_1} & K_{G_2G_2} \end{bmatrix} = \begin{bmatrix} K(\mathbf{s_1}, \mathbf{s_1'}) & \frac{K(\mathbf{s_1}, \mathbf{s_b})K(\mathbf{s_2'}, \mathbf{s_b})}{K(\mathbf{s_b}, \mathbf{s_b})} \\ \frac{K(\mathbf{s_2}, \mathbf{s_b})K(\mathbf{s_1'}, \mathbf{s_b})}{K(\mathbf{s_b}, \mathbf{s_b})} & K(\mathbf{s_2}, \mathbf{s_2'}) \end{bmatrix}. \tag{2}$$

It is then trivial to contribute more cells into this model.

To add complexity to this model, I could either additively or multiplicatively contribute additional interaction terms into the off-diagonals. For example, I could use the spatially inspired kernel [5]

$$\tilde{\mathbf{K}} = \mathbf{K} + C(\theta) + \tau^2 I \qquad C_{ij}(\theta) = \sigma^2 \exp(\frac{-||s_i - s_j||}{l}). \tag{3}$$

Furthermore, I can add expectations for these gene levels to converge to the empirical spatial distributions of gene levels in cell colonies over time. Ultimately, I plan to use these multivariate, multi-index Gaussian Processes to estimate the strength of the interaction terms, determine whether the interaction is additive or multiplicative, and estimate the global influences, potentially on the already existing *E. coli* data provided by van Vliet et al. [4]. In addition, following Sarkans et al., I plan to develop lower bounds for these estimators to address the computational complexity associated with high-dimensional multivariate, multi-index Gaussian Processes [3].

Preliminary Results

I ran a preliminary simulation. In the simulation, a stationary cell at the origin divided into two cells that then traveled through space, with each cell following

$$\vec{r}_1 = vt\hat{x} + A\sin(\omega t)\hat{y} \qquad \vec{r}_2 = vt\hat{x} - A\sin(\omega t)\hat{y}$$

utilizing the additive kernel when defining K. The parameters can be found in Table 1. As demonstrated in Figure 1, as the cells approach each other, their gene levels converge, and, as the cells move away, their gene levels diverge, illustrating that this model is capable of describing spatio-temporal covariances.

References

[1] Alexis Boukouvalas, James Hensman, and Magnus Rattray. Bgp: identifying gene-specific branching dynamics from single-cell data with a branching gaussian process. *Genome biology*, 19(1):65, 2018. 1, 3

[2] Alfredo A Kalaitzis and Neil D Lawrence. A simple approach to ranking differentially expressed gene expression time courses through gaussian process regression. *BMC bioinformatics*, 12(1):180, 2011. 1

[3] Elvijs Sarkans, Sumon Ahmed, Magnus Rattray, and Alexis Boukouvalas. Modelling sequential branching dynamics with a multivariate branching gaussian process. *Transactions on Machine Learning Research*, 2023. 2, 3

[4] Simon van Vliet, Alma Dal Co, Annina R Winkler, Stefanie Spriewald, Barbel Stecher, and Martin Ackermann. Spatially correlated gene expression in bacterial groups: The role of lineage history, spatial gradients, and Cell-Cell interactions. *Cell Systems*, 6(4):496–507, 2018. 1, 3

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A Graphs and Tables

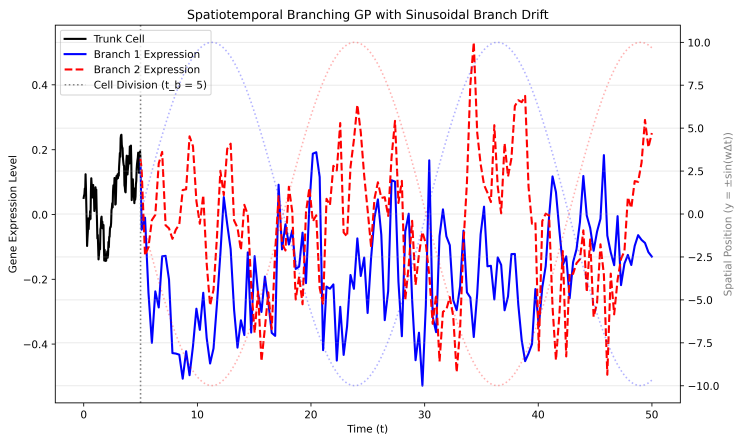

Figure 1: Sample simulation of post-branch cell division with spatial correlations between cells with an additive kernel. The translucent lines in the background are indicative of the position of the two branch cells.

Category	**Parameter**	**Value**
Temporal	σ_t	0.05
Temporal	γ_t	0.01
Spatial (x)	σ_x	0.125
Spatial (x)	γ_x	1
Spatial (y)	σ_y	0.125
Spatial (y)	γ_y	1
Movement	v (velocity in x)	0.5
Movement	ω (angular frequency in y)	0.25
Movement	A (amplitude in y)	10

Table 1: Model parameters for temporal, spatial, and movement components.